

DATA ANALYSIS REPORT

Healthcare Data Analysis & Interactive Business Intelligence Dashboard

*Analyzing Patient Demographics, Admissions, Diagnoses, Billing,
Insurance and Hospital Performance*

Daniel Sowah

Data Science and Analytics

Visualization and Story Telling Using Power BI

2025

Executive Summary

This report presents a data analysis and business intelligence project focused on healthcare operations and hospital performance. The analysis uses a hospital dataset obtained from Kaggle containing patient demographics, medical conditions, admission and discharge information, physician details, insurance providers, billing amounts, medications and test results. Python was used for data cleaning and preparation, while Power BI was used to model, analyze and visualize the data through interactive dashboards.

The analysis identified important patterns in patient volume, admission types, diagnoses, billing, insurance payments, discharge activity, length of stay and provider performance. The dataset contains 25,000 patients, with an average billing amount of approximately \$26,000 and an average hospital stay of 15.51 days. Arthritis was the most frequently recorded medical condition, with approximately 4.4 thousand cases and \$112 million in billing. Elective, emergency and urgent admissions were relatively balanced, while discharge activity peaked in 2020 and declined in 2024, which the source report attributes likely to incomplete-year data.

The resulting dashboards are intended to support evidence-based decision-making around resource allocation, patient flow, clinical prioritization, billing, insurance management and operational performance.

1. Introduction

1.1 Background

Modern healthcare organizations generate large volumes of data from patient demographics, admission records, treatment outcomes, billing and insurance information. The source paper highlights that these data are often dispersed across systems, captured inconsistently or not fully utilized, making real-time decision-making difficult. Business Intelligence and data visualization provide a means of converting this complex information into accessible insights.

1.2 Problem Statement

Hospitals need to balance operational efficiency, quality of patient care and financial sustainability. Fragmented patient, clinical and financial data can make it difficult to monitor treatment outcomes, manage resources, understand patient flow, improve billing and support discharge planning.

1.3 Aim

The aim of the project is to employ data analysis and interactive visualization to identify trends and problems in patient admissions, billing performance and treatment outcomes, and to present the results through an understandable hospital performance dashboard.

1.4 Objectives

1. Clean and prepare the hospital dataset for analysis.
2. Analyze patient demographic characteristics and medical conditions.
3. Examine patient admission and discharge patterns.
4. Analyze billing and insurance performance.
5. Evaluate hospital and doctor performance.
6. Develop interactive dashboards using Power BI and DAX.
7. Generate insights and actionable recommendations for healthcare decision-making.

1.5 Business Questions

- What are the major patterns in patient volume and demographics?
- Which medical conditions occur most frequently and generate the greatest billing?
- How are elective, emergency and urgent admissions distributed?

- How do admission and discharge patterns change over time?
- What relationship exists between length of stay and billing?
- How do insurance providers contribute to billing?
- How are hospital and doctor workloads distributed?

2. Dataset Description

2.1 Data Source

The source paper states that the hospital dataset was obtained from Kaggle. It contains a mixture of numerical and categorical variables covering patients, medical information, billing, admissions and physicians.

2.2 Dataset Overview

Data Area	Variables / Information Included
Patient information	Patient name, age, gender, blood type
Clinical information	Medical condition, medication, test results
Admission information	Admission date, discharge date, admission type
Provider information	Doctor and hospital information
Financial information	Billing amount, insurance provider

2.3 Data Dictionary – Key Fields

Field	Purpose
Patient name	Identifies the patient record in the source dataset
Age	Supports demographic and segmentation analysis
Gender	Supports gender-based comparisons
Medical condition	Identifies diagnosed conditions
Admission date	Supports admission trend analysis
Discharge date	Supports discharge and length-of-stay analysis
Doctor	Supports provider workload and performance analysis
Billing amount	Supports financial analysis
Insurance provider	Supports insurance payment analysis
Admission type	Categorizes elective, emergency and urgent admissions
Medication	Provides treatment-related information
Test results	Provides admission/discharge test-result information

2.4 Data Quality Assessment

The source paper reports that the raw dataset required cleaning because of duplicate patient names, blank or unusual values in Medical Conditions, Test Results and Medications, and missing values in the Admission Date field. These issues were addressed before dashboard development.

3. Data Cleaning and Preparation

3.1 Data Cleaning

- Python was used to remove duplicate entries in patient names.
- Blank and unusual values in Medical Conditions, Test Results and Medications were addressed.
- Missing values in the Admission Date field were handled.
- The dataset was prepared for subsequent Power BI analysis.

3.2 Data Transformation and Modelling

The cleaned data was imported into Power BI. Category fields were created to support the conceptual model. DAX measures were then developed for key analytical metrics.

3.3 DAX Analysis

Measure / Calculation	Analytical Purpose
Total number of patients	Measures overall patient volume
Average billing amount	Measures average financial activity per patient
Total number of doctors	Measures provider count
Diagnosis count	Measures frequency of medical conditions
Doctor workload	Supports provider workload comparison
Days spent in hospital	Supports length-of-stay analysis

3.4 Analysis Workflow

Raw hospital data → Python cleaning → data preparation → Power BI data model → DAX measures → interactive dashboards → analysis and insights → recommendations

4. Exploratory Data Analysis

4.1 Descriptive Statistics and KPIs

KPI	Reported Result
Total patients	25,110
Average Age	51,8
Average billing per Patients	\$26,879
Average length of stay	15.5days
Total Medical conditions	6
Total Admissions	2,192

Readmission Rate	8,7%
Top Medications	5

4.2 Patient Demographics

The dashboard overview reports 25,000 patients, of whom approximately 11,000 were female. The medical-condition analysis also reports higher male patient counts than female counts across the displayed medical-condition categories.

4.3 Diagnosis Analysis

Arthritis was the most frequently recorded condition, with approximately 4.4 thousand cases. Asthma was the lowest among the six conditions referenced in the source analysis, with approximately 3.9 thousand cases. Arthritis also generated approximately \$112 million in billing.

4.4 Admission and Discharge Analysis

Monthly admissions were generally around 2,000 per month. March and August recorded the highest reported admission counts, at 2,205 and 2,218 respectively. Elective, emergency and urgent admissions were relatively balanced, with approximately 8,128–8,510 records across the three categories. Discharges peaked in 2020 at approximately 5.2 thousand, remained around 5 thousand per year, and fell to approximately 1.7 thousand in 2024; the source report notes that the 2024 decline may reflect incomplete-year data.

4.5 Financial and Insurance Analysis

Billing was relatively consistent across the displayed doctors, with John Smith showing a slight lead at approximately \$0.35 million in the referenced chart. The analysis identifies a positive relationship between longer hospital stays and higher billing. UnitedHealthcare and Medicare accounted for the largest share of insurance payments in the dashboard analysis, with total insurance billing reported at \$5.08 million. Total billing peaked in 2020 at \$132 million, remained around \$128 million from 2021 to 2023, and declined to \$45 million in 2024, which the source again attributes to incomplete data.

5. Advanced Analysis

5.1 Trend Analysis

Trend analysis was used to examine monthly admissions, discharge activity and billing over time. The analysis highlights the 2020 discharge and billing peaks and the apparent 2024 decline.

5.2 Relationship Between Length of Stay and Billing

The line chart in the original analysis indicates a positive correlation between longer hospital stays and higher billing. This relationship is important for operational monitoring because length of stay can influence both capacity utilization and financial activity.

5.3 Segmentation Analysis

The dashboard uses categories such as gender, admission type, medical condition and insurance provider to compare patient groups and financial activity. These segments help management identify differences in patient volume, diagnoses and billing.

5.4 Provider and Hospital Performance

The hospital-level analysis reported Willian's Ltd as the leading hospital by patient volume in the displayed chart, with 23 patients. Other leading hospitals ranged from 14 to 21 patients. Doctor workloads were described as relatively evenly distributed, and billing values were tightly clustered.

6. Dashboard Development

6.1 Dashboard Objectives

The project developed four dashboard areas: Hospital Performance Overview; Patient Admission & Discharge Insights; Medical Conditions & Treatment Overview; Financial Analysis & Insurance Insights; and Hospital & Doctor Performance. Together these views provide an interactive picture of clinical, operational and financial performance.

6.2 Visualization Selection

Visual	Purpose	Strength	Limitation
Line chart	Trends over time	Shows patterns and changes	Can become difficult with many lines
Bar/column chart	Category comparison	Simple and direct	Less suitable for very large category sets
Stacked bar chart	Distribution within categories	Shows composition	Can become cluttered
Pie/donut chart	Parts of a whole	Easy to interpret proportions	Less precise for detailed comparisons
Matrix/table	Detailed KPI view	Readable detail	Less visually appealing
KPI card	Key metrics	Highlights critical figures	Limited context alone

6.3 Dashboard KPI Design

- Total patients
- Average billing per Patient
- Average length of stay
- Admission counts
- Diagnosis counts
- Insurance billing
- Bed Occupancy Rate
- Average Age
- Readmission Rate
- Highest billing
- Insured Patients

6.4 Dashboard Pages

8. Hospital Performance Overview – summarizes patient volume, average billing, average stay, gender, diagnoses and admission trends.
9. Patient Admission & Discharge Insights – examines monthly admissions, admission types, length of stay and discharge trends.
10. Medical Condition & Treatment Overview – compares conditions by gender, billing and test results.
11. Financial Analysis & Insurance Insights – examines billing, insurance contributions, billing and billing trends.
12. Patient Demographics – compares patient volume, workload, billing and discharge activity.

7. Key Findings and Insights

13. High Patient Volume: The hospital dataset contains more than 25,110 patient records, with female patients representing approximately 44%.
14. Common Condition: Arthritis is the most frequently diagnosed condition, with about 4.4k cases and approximately \$112M in billing.
15. Balanced Admission Types: Elective, emergency and urgent admissions are relatively balanced at approximately 8.1k–8.5k each.
16. Discharge Trend: Discharges peaked in 2020 and then remained relatively stable before a marked 2024 decline that may reflect incomplete data.
17. Gender Difference: The dashboard reports more male than female patients across the displayed medical-condition categories.
18. Billing Relationship: The analysis indicates that longer hospital stays are associated with higher billing.
19. Insurance Impact: UnitedHealthcare and Medicare account for the largest share of insurance payments in the displayed analysis.
20. Provider Performance: Doctor billing is relatively consistent, with John Smith showing a slight lead, while hospital and doctor workloads appear broadly distributed.

8. Business Recommendations

21. Target high-cost conditions by developing care-management approaches for conditions such as Arthritis.
22. Review gender-specific diagnosis differences and investigate whether targeted health interventions are appropriate.
23. Optimize length of stay by reviewing patient pathways and identifying opportunities to improve efficiency without compromising care quality.
24. Enhance discharge efficiency by monitoring discharge congestion and planning smoother transitions during peak periods.
25. Maintain standardized billing practices across doctors and departments to support transparency and compliance.
26. Leverage insurance data when negotiating with major insurers by using evidence on billing contributions and patient volumes.
27. Use real-time dashboards so department leaders can monitor daily operations and make evidence-based decisions.

9. Limitations

- The 2024 decline in admissions/discharges and billing may be affected by incomplete-year data, as noted in the source paper.
- The analysis is dependent on the quality and completeness of the original Kaggle dataset.
- The reported relationship between length of stay and billing should be interpreted as an observed correlation rather than proof of causation.
- The source paper describes dashboard-level findings but does not provide a complete statistical testing framework for all relationships.
- The dataset represents the selected hospital dataset and therefore should not automatically be generalized to all healthcare organizations.

10. Conclusion

The project demonstrates how healthcare data can be transformed into a practical Business Intelligence solution using Python, Power BI and DAX. The resulting dashboards provide a data-driven view of patient volume, average billing, length of stay, gender distribution, medical conditions, admission types, discharge activity and insurance billing.

The analysis identifies Arthritis as a high-frequency and high-billing condition, highlights relatively balanced admission types, shows a relationship between longer stays and higher billing, and identifies important changes in discharge and billing trends over time. These findings can support resource planning, patient-flow improvement, clinical prioritization, billing management and operational performance monitoring.

11. Future Improvements

- Introduce predictive forecasting for admissions, discharges and billing.
- Add automated or scheduled data refresh for operational monitoring.
- Integrate additional hospital and clinical datasets.
- Develop anomaly detection for unusual billing, admissions or length-of-stay patterns.
- Extend the dashboard with more granular department-level analysis.
- Evaluate the relationships identified in the dashboard using formal statistical tests.

12. Technical Implementation

12.1 Tools and Technologies

Technology	Role in Project
Python	Data cleaning and preparation
Power BI	Data modelling, dashboard development and interactive visualization
DAX	Analytical measures and calculations
Kaggle	Source of the hospital dataset

12.2 End-to-End Workflow

Kaggle hospital dataset → Python cleaning → data preparation → Power BI model → DAX measures → dashboard development → exploratory and trend analysis → insights → recommendations

12.3 Power BI Justification

Power BI was selected because it integrates well with Microsoft Office products and Excel, offers a user-friendly drag-and-drop interface, supports filters, drill-downs and interactive queries, and provides Power BI Service features including role-based access and compliance capabilities. The source paper also describes it as cost-effective for many healthcare institutions.

13. References

28. Belle, A., Thiagarajan, R., Soroushmehr, S. M. R., Navidi, F., Beard, D. A. & Najarian, K. (2015) 'Big data analytics in healthcare', BioMed Research International, 2015, pp. 1–16. Available at: <https://doi.org/10.1155/2015/370194>

29. Davenport, T. H. & Harris, J. G. (2007) *Competing on analytics: The new science of winning*. Boston: Harvard Business School Press.
30. Few, S. (2009) *Now you see it: Simple visualization techniques for quantitative analysis*. Oakland, CA: Analytics Press.
31. Healthcare Information and Management Systems Society (HIMSS) (2020) 'Case studies in healthcare analytics adoption', HIMSS.org.
32. Inmon, W. H., Strauss, D. & Neushloss, G. (2010) *DW 2.0: The architecture for the next generation of data warehousing*. Amsterdam: Elsevier.
33. Kankanhalli, A., Hahn, J. & Tan, S. S.-L. (2016) 'Business intelligence in healthcare: A study of antecedents and consequences', *Information Systems Frontiers*, 18(2), pp. 257–272.
34. Kohli, R. & Tan, S. S.-L. (2016) 'Electronic health records: How can IS researchers contribute to transforming healthcare?', *MIS Quarterly*, 40(3), pp. 553–573.
35. Mehta, N. & Pandit, A. (2018) 'Concurrence of big data analytics and healthcare: A systematic review', *International Journal of Medical Informatics*, 114, pp. 57–65.
36. Raghupathi, W. & Raghupathi, V. (2014) 'Big data analytics in healthcare: Promise and potential', *Health Information Science and Systems*, 2(1), pp. 1–10.
37. Sharma, R., Yetton, P. & Zmud, R. W. (2016) 'Implementation costs of healthcare information systems: A framework and research agenda', *Information Systems Research*, 27(4), pp. 767–783.
38. Wang, Y., Kung, L., Wang, W. Y. C. & Cegielski, C. G. (2018) 'An integrated big data analytics-enabled transformation model: Application to health care', *Information & Management*, 55(1), pp. 64–79.
39. Kaggle (n.d.) Healthcare Dataset. Available at:
<https://www.kaggle.com/datasets/prasad22/healthcare-dataset>

14. Appendices

Appendix A – Key Dashboard Metrics

Metric	Value from Source Analysis
Patients	25,110
Female patients	11,000 / 44%
Average billing	\$26k
Average stay	15.51 days
Arthritis	4.4k cases / \$112M billing
2020 total billing	\$132M
2021–2023 total billing	Approximately \$128M per year
2024 total billing	\$45M (source notes incomplete data)
Total insurance billing	\$5.08M

Appendix B – Dashboard Evidence

The original paper contains the dashboard wireframes/figures for Hospital Performance Overview, Patient Admission & Discharge Insights, Medical Condition & Treatment Overview, Financial Analysis & Insurance Insights, and Hospital & Doctor Performance. Those figures can be inserted here when the original dashboard image files are available.

Appendix C – Original Project Context

Original assignment title: Report on Designing a business dashboard in Power BI for a chosen organization. Programme title: Visualization and Story Telling Using Power BI. Name: Daniel Sowah. Year: 2025.